Practical 1

Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

1. Problem Definition

Project Name: StudentHub Portal

StudentHub is a web-based student portal designed to provide students and faculty with a centralized platform for accessing academic information and student-related services.

The portal will provide different features based on user roles such as Student, Faculty, and Administrator. The system will include modules for dashboard, profile, courses, attendance, assignments, timetable, notices, results, events, and communication.

The objective of this project is to create a structured foundation for the semester-long development of the StudentHub portal.

2. User Roles

2.1 Student

- Login to the portal
- View dashboard
- Manage profile
- View enrolled courses
- Check attendance
- View assignments
- View timetable
- View results
- View notices and events

2.2 Faculty

- Login to the portal
- View faculty dashboard
- Manage profile
- Manage courses
- Upload assignments
- Manage attendance
- Publish notices
- View student information

2.3 Administrator

- Manage students and faculty
- Manage courses
- Manage notices

- Manage events
- Manage system information
- Monitor the overall portal

3. Functional Requirements

General System Requirements: The system shall provide a login page for users, authenticate users using their login credentials, identify users based on their assigned role, provide a separate dashboard according to the user's role, provide navigation between all major pages, provide a logout option, and allow users to view important notices and upcoming college events.

Student Requirements: Students shall be able to access their dashboard, view and update their profile, view their enrolled courses, view their attendance, view their assignments, view their timetable, view their examination results, view important notices, and view upcoming college events.

Faculty Requirements: Faculty shall be able to access the faculty dashboard, manage student attendance, manage assignments, manage course information, view important notices, and view upcoming college events.

Administrator Requirements: Administrators shall be able to access the admin dashboard, manage users, manage course information, manage notices, manage college events, and manage overall portal information.

4. Minimum Pages

1. Home Page
2. Login Page
3. Student Dashboard
4. Student Profile
5. Courses Page
6. Attendance Page
7. Assignments Page
8. Timetable Page
9. Results Page
10. Notices Page
11. Events Page
12. Faculty Dashboard
13. Faculty Courses Page
14. Faculty Attendance Page
15. Faculty Assignments Page
16. Admin Dashboard

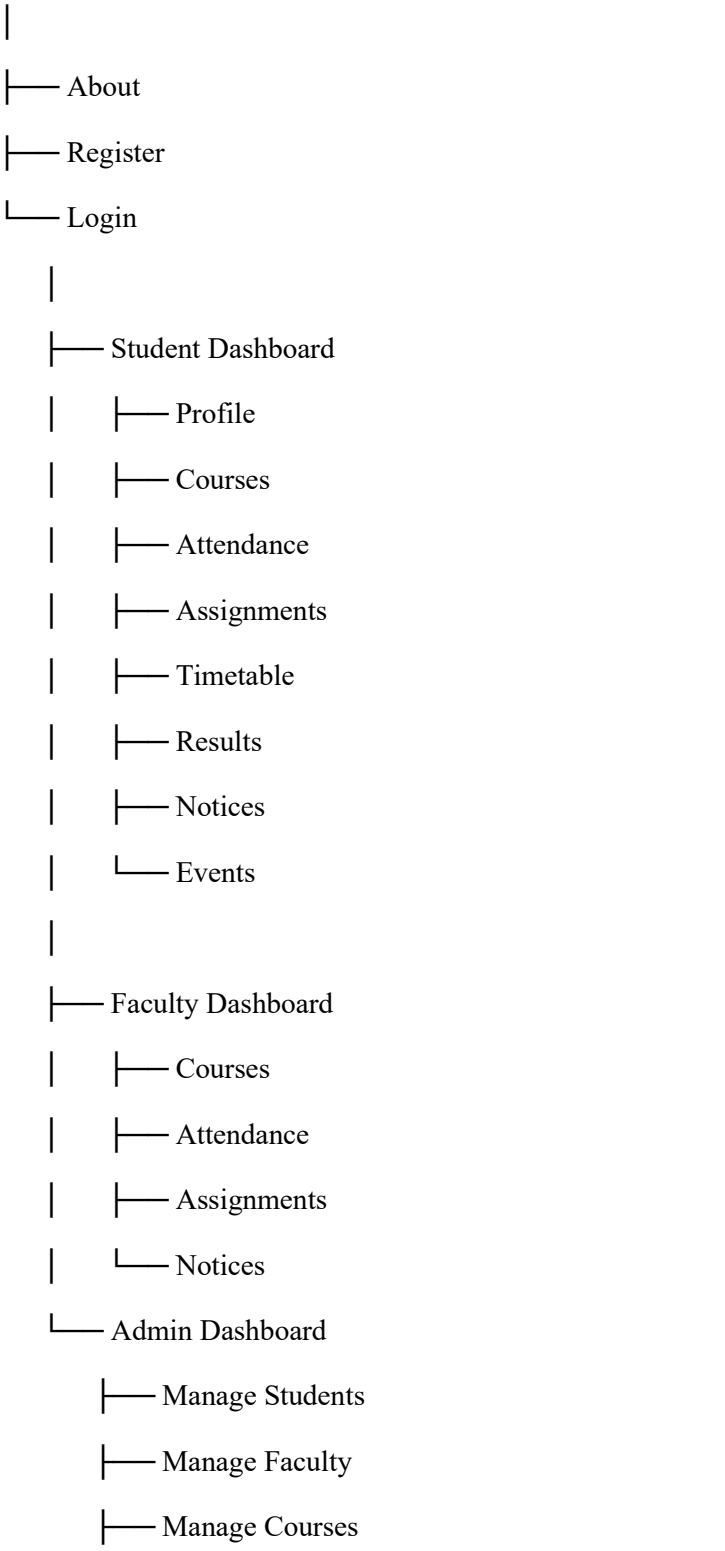
5. Sitemap

StudentHub Portal

- Home
- Login
- Student
  - Dashboard
  - Profile
  - Courses
  - Attendance
  - Assignments
  - Timetable
  - Results
  - Notices
  - Events
- Faculty
  - Dashboard
  - Courses
  - Attendance
  - Assignments
  - Notices
- Administrator
  - Dashboard
  - Manage Students
  - Manage Faculty
  - Manage Courses
  - Manage Notices
  - Manage Events

6. Navigation Flow

Home



├── Manage Notices

└── Manage Events

Navigation between HTML pages will be managed using HTML hyperlinks (<a href="">). A common navigation bar can be included on all pages so users can easily move between related pages.

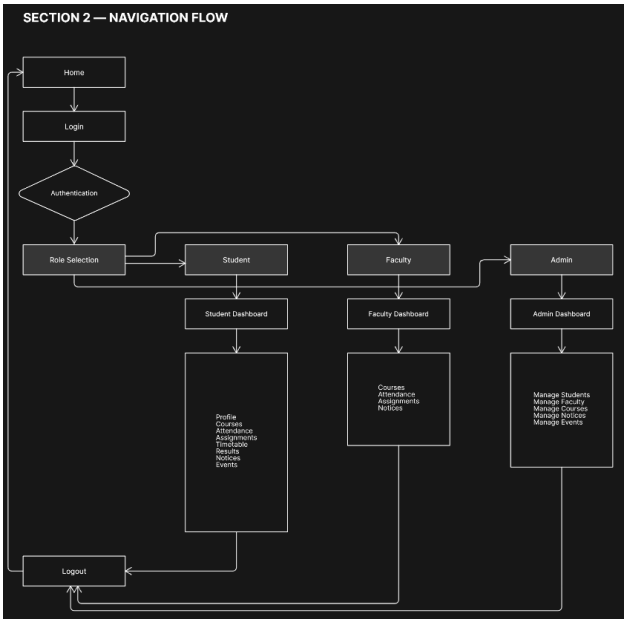
Dashboard

Courses

Attendance

Assignments

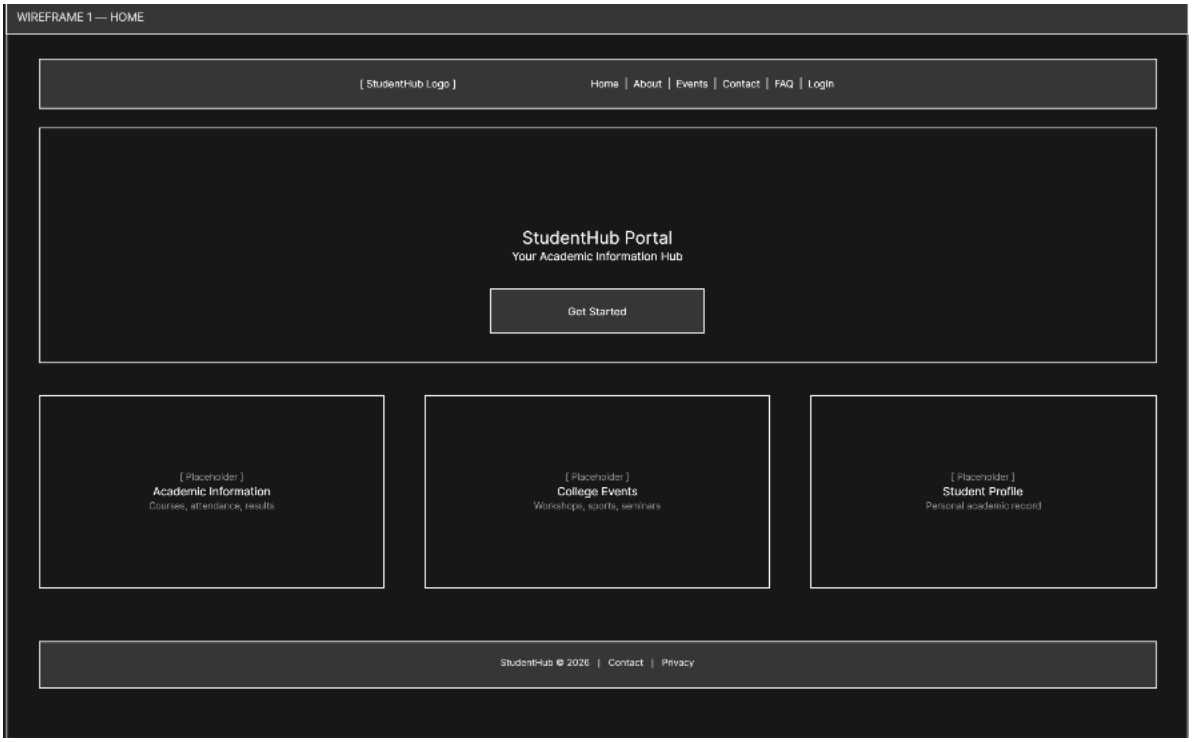
Logout



7. Low-Fidelity Wireframe

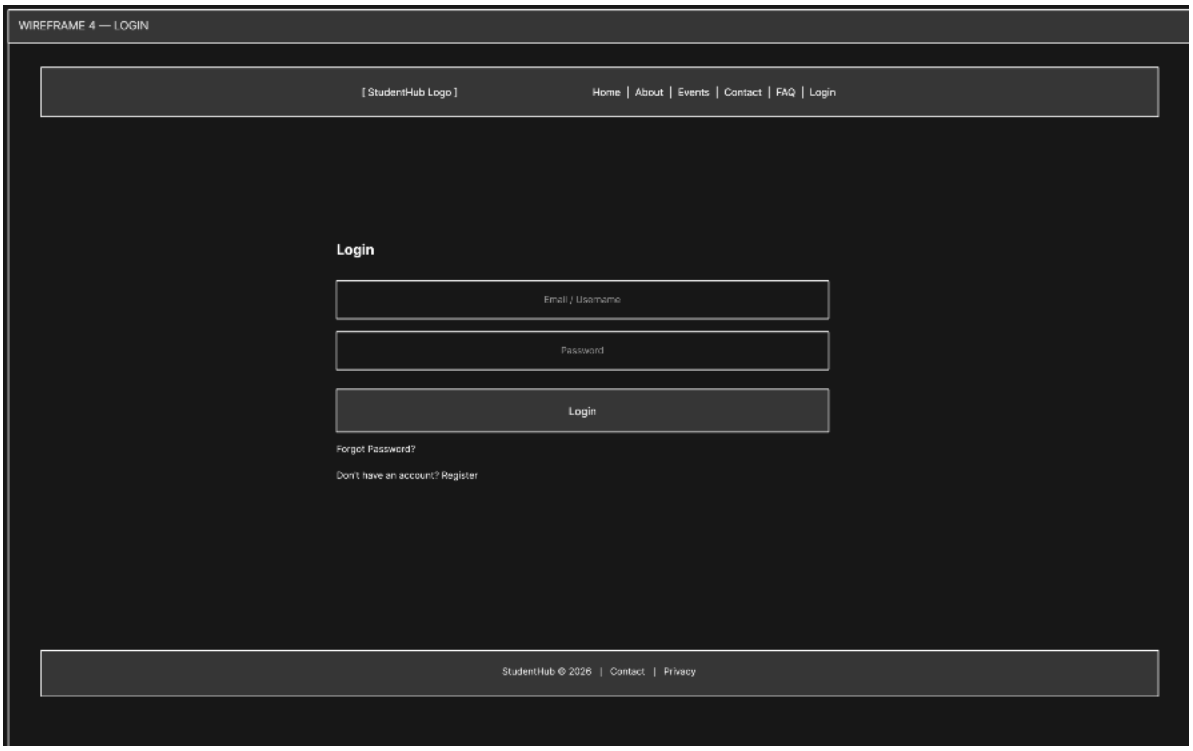
7.1 Home Page

Area	Content
Header	**STUDENTHUB**
Navigation	Home | About | Register | Login
Main Heading	**STUDENTHUB PORTAL**
Description	Your Academic Information Hub
Main Button	**[Get Started]**
Features Section	Academic Information | Events | Student Profile
Footer	About | Features | Contact



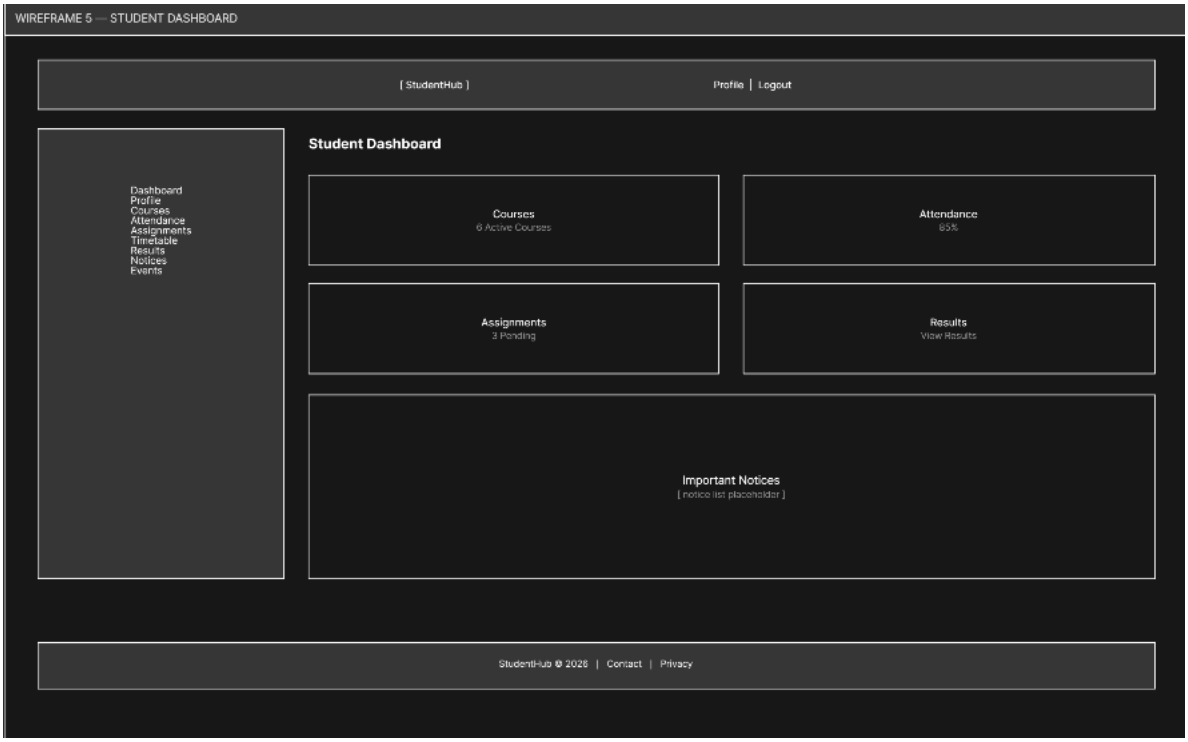
7.2 Login Page

Area	Content
Header	**STUDENTHUB**
Page Heading	**Login**
Username Field	Email / Username
Password Field	Password
Action	**[Login]**
Additional Link	Don't have an account? **Register**
Footer	StudentHub © 2026



7.3 Student Dashboard

Area	Content
Header	**STUDENTHUB** | Profile | Logout
Sidebar	Dashboard
Sidebar	Profile
Sidebar	Courses
Sidebar	Attendance
Sidebar	Assignments
Sidebar	Timetable
Sidebar	Results
Sidebar	Notices
Sidebar	Events
Main Content	**Student Dashboard**
Dashboard Card 1	Courses — **6**
Dashboard Card 2	Attendance — **85%**
Dashboard Card 3	Pending Assignments — **3**
Dashboard Card 4	Results — **View**
Notice Area	Important Notices



7.4 Courses Page

Area	Content
Header	**STUDENTHUB** | Profile | Logout
Sidebar	Dashboard
Sidebar	Profile
Sidebar	Courses
Sidebar	Attendance
Sidebar	Assignments
Sidebar	Timetable
Sidebar	Results
Main Heading	**My Courses**
Table Column 1	Course Name

Area	Content
Table Column 2	Faculty
Table Column 3	Credits
Course 1	Data Structures
Course 2	Web Technology
Course 3	Python
Course 4	Artificial Intelligence

7.5 Attendance Page

Area	Content
Header	**STUDENTHUB** | Profile | Logout
Sidebar	Dashboard
Sidebar	Profile
Sidebar	Courses
Sidebar	Attendance
Sidebar	Assignments
Sidebar	Timetable
Sidebar	Results
Main Heading	**Attendance**

Area	Content
Table Column 1 Course	
Table Column 2 Attendance	
Data 1	Data Structures — **85%**
Data 2	Web Technology — **90%**
Data 3	Python — **82%**
Data 4	AI/ML — **88%**

8. Project Folder Structure

```
StudentHub/
│
├── index.html
├── README.md
│
├── pages/
│   ├── login.html
│   ├── student-dashboard.html
│   ├── profile.html
│   ├── courses.html
│   ├── attendance.html
│   ├── assignments.html
│   ├── timetable.html
│   ├── results.html
│   ├── notices.html
│   ├── events.html
│   ├── faculty-dashboard.html
│   ├── faculty-courses.html
│   ├── faculty-attendance.html
│   ├── faculty-assignments.html
│   └── admin-dashboard.html
│
├── css/
│   └── style.css
│
├── js/
│   └── script.js
│
├── images/
```

```
│   └── README.txt
│
└── docs/
    ├── sitemap.png
    └── wireframe.png
```

9. GitHub Commit Plan

Practical	Commit Message
Practical 1	Initial project setup and requirement analysis
Practical 2	Added basic HTML structure and navigation
Practical 3	Added CSS styling and responsive layout
Practical 4	Added JavaScript functionality
Practical 5	Added forms and validation
Practical 6	Updated student modules
Practical 7	Updated faculty modules
Practical 8	Updated admin modules
Final	Final StudentHub portal implementation

Basic Git Commands

git init
git add .
git commit -m "Initial project setup and requirement analysis"
git branch -M main
git remote add origin <GitHub-Repository-URL>
git push -u origin main

For subsequent practicals:
git add .
git commit -m "Description of changes"
git push

10. README.md

StudentHub Portal

StudentHub is a web-based student portal developed as a semester-long academic project.

Project Objective

The objective of StudentHub is to provide students, faculty, and administrators with a centralized platform for managing and accessing academic information.

User Roles

Student

- View dashboard
- Manage profile
- View courses
- Check attendance
- View assignments
- View timetable
- View results
- View notices and events

Faculty

- Manage courses

- Manage attendance

- Manage assignments

- Publish notices

- View student information

Administrator

- Manage students

- Manage faculty

- Manage courses

- Manage notices

- Manage events

Main Modules

- Authentication

- Student Dashboard

- Faculty Dashboard

- Admin Dashboard

- Profile Management

- Course Management

- Attendance

- Assignments

- Timetable

- Results

- Notices

- Events

Technologies

- HTML5
- CSS3
- JavaScript
- Git
- GitHub

Project Structure

StudentHub/

- index.html
- pages/
- css/
- js/
- images/
- docs/
- README.md

Navigation

HTML hyperlinks are used to navigate between different pages of the StudentHub portal.

Future Scope

The project can be extended with:

- Database integration
- User authentication
- REST APIs
- Online assignment submission
- Notifications
- Attendance management
- Result management

- ER diagram and backend implementation

Author

StudentHub Semester Project

11. Answers to Key Questions

Q1. What is URL and what are its parts?

URL (Uniform Resource Locator) is the address used to locate a resource on the Internet.

Example: https://www.example.com/studenthub/login.html?id=101#profile

- https → Protocol/Scheme
- www.example.com → Domain name
- /studenthub/login.html → Path
- ?id=101 → Query parameter
- #profile → Fragment

Q2. How is an HTML file processed in a web browser?

HTML File
↓
Browser Reads HTML
↓
HTML Parser
↓
DOM (Document Object Model)
↓
CSS + JavaScript Processing
↓
Rendering
↓
Web Page Displayed

The browser reads the HTML document and converts it into a DOM tree. It then processes CSS and JavaScript and finally renders the page on the screen.

Q3. How will page navigation flow be managed among all HTML pages?

Page navigation will be managed using HTML anchor (<a>) tags.

Courses

When the user clicks the link, the browser loads courses.html. A common navigation bar will be provided across the pages to allow users to move between Dashboard, Profile, Courses, Attendance, Assignments, Timetable, Results, Notices, and Events.

Q4. How will GitHub commits be maintained after each practical?

After completing every practical, the updated project files will be added and committed to Git.

```
git add .
git commit -m "Description of changes"
git push
```

Each practical will have a meaningful commit message so that the development history can be tracked easily.

Example:

Practical 1 → Initial project setup and requirement analysis
Practical 2 → Added basic HTML structure and navigation
Practical 3 → Added CSS styling
Practical 4 → Added JavaScript functionality

Github Repository Link: - https://github.com/pateltej2087-pro/STUDENTHUB

Github Screenshot: -

